

LewyGym: A Parkinson's Disease Protein Variant Effect Benchmark

Reveals Signal Inversion in Conservation-Based Protein Language Models

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GitHub: github.com/Tyronita/LewyGym * HuggingFace: [EvanOLeary/LewyGym](https://huggingface.co/EvanOLeary/LewyGym) * License: CC0

Protein language models (PLMs) trained on evolutionary sequence conservation achieve strong variant effect prediction, with ESM-2 650M reaching mean Spearman rho +0.419 across ProteinGym's 217-assay substitution benchmark. However, no benchmark evaluates PLMs across a disease-coherent Parkinson's disease (PD) gene panel. PD spans two opposing pathogenic mechanisms: gain-of-function (GOF) aggregation (SNCA, LRRK2) and loss-of-function (LOF) enzyme/kinase disruption (GBA, PRKN, PINK1, PARK7). We introduce LewyGym, a ProteinGym-compatible benchmark covering seven PD genes. LewyGym provides two tracks: (i) a DMS fitness track with 12,860 missense variants across five alpha-synuclein (SNCA) assays from MaveDB, and (ii) a ClinVar pathogenicity classification track with 421 annotated missense variants across seven PD genes. Baseline ESM-2 evaluation yields mean Spearman rho -0.139 to -0.171 on the DMS track -- a 0.58-unit inversion relative to ProteinGym. All five assays are negative; larger models amplify the inversion. LewyGym is released under CC0 with a reproducible scoring script.

Keywords: protein language model, Parkinson's disease, variant effect prediction, benchmark, alpha-synuclein, deep mutational scanning, gain-of-function

1. Introduction

Protein language models (PLMs) trained on large protein sequence corpora have demonstrated remarkable zero-shot ability to predict the functional consequences of amino acid substitutions [1]. The dominant evaluation paradigm -- exemplified by ProteinGym [1] and FLIP [2] -- measures Spearman rank correlation between model-predicted log-likelihood ratios and experimentally measured fitness scores from deep mutational scanning (DMS) assays. ESM-2 650M achieves a mean Spearman rho of +0.419 across ProteinGym's 217 substitution assays [1], and ESM-1v reaches +0.440 [3].

This strong performance rests on a key biological assumption: amino acids disfavoured by evolution (low PLM log-likelihood) are also functionally damaging. For loss-of-function (LOF) proteins -- where variants disrupt enzymatic activity, structural stability, or binding affinity -- this assumption holds. The evolutionary record penalises variants that break the protein.

Parkinson's disease (PD) presents a systematic challenge to this assumption. The PD gene panel spans two opposing pathogenic mechanisms. SNCA (alpha-synuclein) and LRRK2 operate through gain-of-function (GOF): variants that increase alpha-synuclein's aggregation propensity, or hyperactivate LRRK2 kinase, are pathogenic because they enhance a toxic function. For GOF proteins, a residue that promotes aggregation may be evolutionarily disfavoured -- meaning conservation-based PLMs assign it a negative log-likelihood delta,

predicting it as evolutionarily unusual, while the DMS fitness score registers harm via a completely different biological mechanism (aggregation-driven toxicity). For LOF genes in the PD panel (GBA, PRKN, PINK1, PARK7), the standard conservation-fitness alignment should hold.

Despite PD affecting more than 10 million people worldwide, no benchmark evaluates PLMs across a disease-coherent PD gene panel spanning both GOF and LOF mechanisms simultaneously. ProteinGym (217 assays) includes one SNCA assay (Newberry 2020, benchmarked as a fitness predictor without disease framing) and one Parkin assay, but five PD genes -- LRRK2, GBA, PINK1, PARK7, VPS35 -- are entirely absent, and no clinical pathogenicity classification track exists. Kryukov et al. [10] previously showed GOF proteins are a general weak spot for variant effect predictors; we instantiate this in the PD context and provide a reusable benchmark infrastructure. We introduce LewyGym -- named after Lewy bodies, the misfolded alpha-synuclein aggregates that define PD pathology. Our baseline results confirm that ESM-2 conservation signals invert on SNCA (mean Spearman rho = -0.139 to -0.171), and that larger ESM-2 models amplify rather than correct this inversion.

2. Benchmark Design

2.1 DMS Fitness Track

The DMS track comprises five SNCA assays from MaveDB [7], all measuring yeast fitness as a proxy for alpha-synuclein membrane toxicity. Newberry et al. [5] measured fitness of all single-residue substitutions in a membrane-targeted GFP fusion construct expressed in yeast. Noh and Newberry [6] measured yeast fitness at four galactose induction concentrations (1%, 0.1%, 0.01%, 0.001%), capturing the concentration-dependent toxicity landscape of alpha-synuclein. All five assays cover the full SNCA sequence (UniProt P37840, L = 140) and originate from the Newberry lab at UCSF. Data were retrieved from MaveDB via the public REST API (api.mavedb.org), converted from HGVS notation to ProteinGym one-letter substitution format, and normalised to zero-mean unit-variance. Synonymous variants (wildtype residue = mutant residue), stop codons, insertions, and reference-mismatch variants were excluded; reported n values reflect missense-only variants. Note that the yeast membrane toxicity assay captures membrane-binding GOF; variants that promote fibrillation but reduce membrane binding (notably A30P, G51D) are known to score as fitness-rescuing in this system despite being familial PD mutations.

Assay ID	n	Phenotype	Source (MaveDB)
SNCA_HUMAN_Newberry_2020	2,588	GFP membrane toxicity (yeast)	00000045-k-1
SNCA_HUMAN_Noh_2026_1pct	2,585	Fitness, 1% galactose induction	00001249-a-1
SNCA_HUMAN_Noh_2026_01pct	2,593	Fitness, 0.1% galactose	00001249-a-2
SNCA_HUMAN_Noh_2026_001pct	2,591	Fitness, 0.01% galactose	00001249-a-3
SNCA_HUMAN_Noh_2026_0001pct	2,503	Fitness, 0.001% galactose	00001249-a-4
Total	12,860		

2.2 ClinVar Pathogenicity Track

The pathogenicity track provides ClinVar-classified missense variants for all seven PD genes: SNCA, LRRK2, GBA, PRKN, PINK1, PARK7, and VPS35. Only missense single-nucleotide variants with at least one submitter and no conflicting interpretations are retained. After deduplication on (mutant, label) pairs, 421 variants remain: 314 Pathogenic/Likely Pathogenic (P/LP) and 107 Benign/Likely Benign (B/LB). Table 2

shows per-gene counts. GBA dominates ($n = 240$) due to the large literature on GBA variants in Gaucher disease, the strongest known genetic risk factor for PD. The evaluation metric is AUROC; model scores are negated before ranking (more negative log-likelihood delta = more damaging).

Gene	Pathogenic mechanism	P/LP	B/LB	n
GBA	LOF -- glucocerebrosidase (GD/PD risk)	237	3	240
LRRK2	GOF -- kinase hyperactivation	10	71	81
PRKN	LOF -- ubiquitin E3 ligase	28	14	42
PINK1	LOF -- mitophagy kinase	21	11	32
PARK7	LOF -- oxidative stress sensor	10	3	13
SNCA	GOF -- aggregation (hallmark PD)	5	3	8
VPS35	GOF -- retromer complex dysfunction	3	2	5
Total		314	107	421

2.3 Format and Compatibility

All DMS files follow ProteinGym substitution CSV format (columns: mutant, DMS_score). Wildtype sequences, gene metadata, and assay phenotype descriptions are provided in `reference_files/DMS_substitutions.csv`. The scoring script `score.py` implements masked-marginal scoring (Section 3.1) and accepts any HuggingFace EsmForMaskedLM checkpoint. Dependencies: transformers, torch, pandas, scipy, numpy. The full DMS track can be reproduced from scratch in under 2 minutes on a CPU for ESM-2 8M, and under 30 minutes for ESM-2 650M.

3. Baseline Evaluation

3.1 Masked-Marginal Scoring

We use masked-marginal scoring [3], the standard ProteinGym evaluation protocol. For each position i in the wildtype sequence (1-indexed, excluding special tokens), position i is masked and a single forward pass computes the log-softmax distribution over the amino acid vocabulary. The score for variant $X \rightarrow Y$ at position i is:

$$\text{score}(X \rightarrow Y) = \log p(Y \mid \text{seq with position } i \text{ masked}) - \log p(X \mid \text{seq with position } i \text{ masked})$$

For multi-mutant variants, per-position log-likelihood deltas are summed. Building the position cache requires L forward passes ($L=140$ for SNCA), approximately 1-2 seconds on CPU for ESM-2 8M and 20-30 seconds for ESM-2 650M. Spearman rank correlation between model scores and experimental DMS fitness values is the primary DMS track metric.

3.2 Models Evaluated

We evaluate two ESM-2 checkpoints [4]: 8M parameters (facebook/esm2_t6_8M_UR50D, 6 transformer layers, 320-dim embeddings) and 650M parameters (facebook/esm2_t33_650M_UR50D, 33 layers, 1280-dim). ESM-2 650M is the largest checkpoint that achieves peak ProteinGym performance ($\rho = +0.419$) before diminishing returns at 3B and 15B parameters [1]. Both models use 32-bit float precision and SDPA attention.

3.3 DMS Track Results

Assay	ESM-2 8M	ESM-2 650M	n
SNCA_HUMAN_Newberry_2020	-0.165	-0.210	2,588
SNCA_HUMAN_Noh_2026_1pct	-0.147	-0.168	2,585
SNCA_HUMAN_Noh_2026_01pct	-0.135	-0.160	2,593
SNCA_HUMAN_Noh_2026_001pct	-0.115	-0.146	2,591
SNCA_HUMAN_Noh_2026_0001pct	-0.134	-0.171	2,503
Mean	-0.139	-0.171	12,860

Table 3. Negative Spearman rho values indicate signal inversion: model conservation scores anti-correlate with experimental fitness. For comparison, ESM-2 650M achieves +0.419 mean Spearman rho on ProteinGym's 217-assay substitution benchmark (Notin et al., NeurIPS 2023 Table A5). The 650M model amplifies the inversion vs 8M (-0.171 vs -0.139) across all five assays. n values reflect missense-only variants (synonymous excluded per ProteinGym protocol).

All five SNCA assays yield negative Spearman correlations for both models (Table 3). ESM-2 650M produces a more negative mean correlation than 8M in all five assays, with the largest inversion on the Newberry 2020 membrane toxicity assay ($\rho = -0.210$). The mean DMS track score is -0.139 (8M) and -0.171 (650M), compared to the ProteinGym substitution benchmark baseline of +0.419 for ESM-2 650M [1] -- a departure of 0.58 Spearman units, exceeding the full per-assay variation range reported for ESM-2 on ProteinGym. Note that the yeast system captures membrane-binding GOF; the known fibril-aggregation variants A30P and G51D are exceptions where this assay and clinical pathogenicity diverge.

4. Discussion

The negative Spearman correlations on all five SNCA DMS assays arise from a mechanistic mismatch between conservation-based scoring and GOF pathogenicity. ESM-2 is trained to reconstruct masked residues from evolutionary context: residues conserved across UniRef50 receive high log-likelihood, and unusual substitutions receive low log-likelihood. For SNCA, the positions most critical for aggregation-driven toxicity are evolutionarily constrained. The model correctly identifies them as evolutionarily disfavoured, producing negative log-likelihood deltas for aggregation-promoting variants -- yet in the yeast DMS assay, these same variants reduce fitness (increase toxicity), and the correlation inverts. In contrast, substitutions that disrupt aggregation-prone contacts are evolutionarily unusual but improve yeast fitness, further inverting the signal.

The amplification of the inversion by larger ESM-2 models (650M > 8M) is mechanistically consistent: larger models learn richer conservation representations and express higher confidence in penalising evolutionarily unusual residues. As SNCA's GOF-relevant variants fall precisely in the 'unusual but toxic via aggregation' category, a more confident conservation model produces a stronger inversion signal. This is distinct from the behaviour on ProteinGym, where larger ESM-2 models improve LOF prediction because deeper conservation encoding better captures functional constraint.

These results have three practical implications. First, conservation-based PLMs should not be used as zero-shot GOF pathogenicity predictors without recalibration or task-specific fine-tuning. Second, aggregate benchmarks like ProteinGym -- which necessarily average across heterogeneous proteins -- may obscure systematic failures on GOF disease proteins. Third, disease-specific benchmarks spanning both GOF and LOF genes within a coherent disease panel are essential for characterising model behaviour on clinically actionable variants.

Limitations: LewyGym v0.1 includes DMS data for SNCA only. LRRK2, GBA, and other PD genes lack large-scale publicly available DMS assays. The ClinVar pathogenicity track has class imbalance (GBA, $n=240$,

dominates) and very small variant counts for some GOF genes (SNCA n=8, VPS35 n=5). Planned additions for v0.2 include LRRK2 kinase activity assays (~70 variants curated from literature), GBA enzyme activity data (~50 variants), and baselines for ESM-1v [3], AlphaMissense [8], and Evo2 [9].

5. Data Availability

All data, code, and results are publicly available at github.com/Tyronita/LewyGym and huggingface.co/datasets/EvanOLeary/LewyGym under CC0 (public domain). DMS data are redistributed from MaveDB under their original CC0 licences. ClinVar data are in the public domain. The scoring script `score.py` reproduces all baseline results in Table 3 from the raw CSV files.

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References

- [1] Notin, P. et al. ProteinGym: Large-Scale Benchmarks for Protein Fitness Prediction and Design. NeurIPS Datasets and Benchmarks 2023. doi:10.1101/2023.12.07.570727
- [2] Dallago, C. et al. FLIP: Benchmark tasks in fitness landscape inference for proteins. NeurIPS Datasets and Benchmarks 2022.
- [3] Meier, J. et al. Language models enable zero-shot prediction of the effects of mutations on protein function. NeurIPS 2021. doi:10.1101/2021.07.09.450648
- [4] Lin, Z. et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science 379, 1123-1130 (2023). doi:10.1126/science.ade2574
- [5] Newberry, R.W. et al. Deep mutational scanning reveals the structural basis of alpha-synuclein activity. Nature Chemical Biology 16, 653-659 (2020). doi:10.1038/s41589-020-0480-6
- [6] Noh, D. and Newberry, R.W. Concentration-dependent fitness landscape of alpha-synuclein. Protein Science 2026. doi:10.1002/pro.70456
- [7] Esposito, D. et al. MaveDB: an open-source platform to distribute and interpret data from multiplexed assays of variant effect. Genome Biology 20, 223 (2019). doi:10.1186/s13059-019-1845-6
- [8] Cheng, J. et al. Accurate proteome-wide missense variant effect prediction with AlphaMissense. Science 381, eadg7492 (2023). doi:10.1126/science.adg7492
- [9] Nguyen, E. et al. Sequence modeling and design from molecular to genome scale with Evo 2. bioRxiv 2024. doi:10.1101/2024.02.27.582234
- [10] Kryukov, K. et al. Systematic benchmarking of variant effect predictors reveals class-specific performance differences. Molecular Systems Biology 2023. PMC10407742
- [11] Fraternali, F. et al. Gain-of-function variants are systematically underperformed by conservation-based computational predictors. Nature Communications 2022.